# AI-MD Course Project: Soft-CLIP for Medical Imaging

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## Abstract

Standard CLIP training treats each image-report pair as the only positive match, even though radiology reports from different studies may describe similar findings. We investigate Soft-CLIP for chest X-ray image-report retrieval, adding pseudo-positive supervision derived from report-embedding similarities. Using MIMIC-CXR with 249,018 training and 2,018 validation pairs, we compare three report-embedding models, two strategies for selecting pseudo-positive reports (a fixed count versus a similarity threshold), and different weightings of the soft-supervision term. The best configuration uses projected BiomedVLP-CXR-BERT embeddings of full reports, threshold-based selection, and a moderate soft-loss weight. It improves R@1 from 9.66% to 11.94% for image-to-text retrieval and from 11.25% to 12.14% for text-to-image retrieval. Overall, threshold-based selection is more reliable than a fixed selection count, and soft supervision works best as a moderate addition to the standard loss rather than a replacement for it. Patient-level evaluation also produces substantially higher scores, suggesting that retrieval may partly rely on patient-specific information. These results highlight the importance of pseudo-positive construction and evaluation criteria in medical image-text retrieval.

## 1 Introduction

Comparison with previous examinations is an established component of radiologic interpretation [1]. This motivates the development of retrieval models that surface relevant prior studies or clinically similar cases as potential tools for decision support and case review [2, 3]. CLIP and similar contrastive vision-language models are the dominant approach for multimodal retrieval today. They assume that any two non-matching images and captions are simply unrelated, an assumption that breaks down in radiology, where different studies frequently describe overlapping clinical findings, so many "negative" pairs are, in fact, partially correct matches.

Prior work has proposed soft supervision to relax this assumption, but existing approaches such as SoftCLIP [4] rely on visual-region and tag similarity rather than report-level semantics, while others, such as Ko and Park [5], combine several sources of clinical similarity without isolating which design choices actually drive retrieval performance. This raises the question of how such report-based soft supervision should be constructed in practice, and which design choices, embedding source, pseudo-positive selection strategy, and soft-loss weight, most affect retrieval quality. We study this question by deriving soft targets purely from report-embedding similarity and evaluating these choices for study-level image-text retrieval in chest radiography.

Our contributions are as follows:

- We implement a Soft-CLIP variant with two pseudo-positive selection strategies, fixed top-$K$ and similarity-threshold selection, and compare them directly under a shared evaluation protocol.
- We evaluate three candidate embedding sources and multiple text granularities for constructing the pseudo-positive similarity matrix.
- We design an evaluation protocol to test the robustness of our results, which surfaces evidence that the model partly relies on patient-identifying cues rather than purely clinical content, and we assess how much this affects our results.

A web-based overview of this work is available on the accompanying project website.

## 2 Background

Chest radiography studies are commonly paired with free-text radiology reports describing the observed findings and clinical impression. Large datasets such as MIMIC-CXR provide chest X-ray studies together with their corresponding reports, enabling the development of models that learn from both visual and textual information [6]. Radiology reports also differ from general image captions in their use of specialized clinical terminology, negation, uncertainty, spatial relations, and domain-specific modifiers [7]. In addition, reports from different studies may contain overlapping clinical information, resulting in meaningful textual or clinical similarity between otherwise unmatched samples [5].

Contrastive Language-Image Pre-training (CLIP) learns a shared representation space by increasing the similarity between matched images and texts while decreasing the similarity between unmatched samples [8]. This approach has been adapted to biomedical vision-language learning in methods such as ConVIRT, GLoRIA, and BioViL [9, 10, 7]. These models use paired medical images and reports to learn representations that can support tasks such as image-text retrieval, classification, and phrase grounding. BioViL further shows the value of modelling the semantic and linguistic characteristics of radiology reports using a domain-adapted text encoder [7].

A limitation of the standard CLIP objective is its use of one-hot supervision. Each image is aligned only with its paired text, while the other texts in the batch are treated as negatives. This assumption may be restrictive when multiple samples contain related semantic content. SoftCLIP addresses this limitation by replacing strict one-hot targets with soft targets that reflect similarities between samples in the image and text modalities [4]. The soft targets assign higher probabilities to samples that are more similar, rather than treating every unpaired sample as an equally negative example. SoftCLIP calculates these similarities using local image regions and text tags and trains the model to match the resulting soft-target distributions using KL divergence.

More recent work has investigated soft supervision specifically for medical vision-language learning. Ko and Park propose clinically enhanced dynamic soft labels based on similarities between chest X-ray reports [5]. Their approach combines textual similarity, similarity between extracted clinical labels, and similarity between report-derived medical graphs. Similarities below predefined thresholds are removed before constructing the soft-label distributions. The method also introduces negation-based hard negatives and graphical alignment to improve the representation of clinical language.

Our project builds on these studies by applying report-based semantic soft supervision to CLIP training on MIMIC-CXR. Similar to SoftCLIP, we relax the assumption that all unmatched pairs should be treated as complete negatives. However, rather than constructing the soft targets from image regions and associated tags, we derive them from semantic similarities between radiology reports generated by an external biomedical language model. For each study, semantically related reports are selected as pseudo-positive targets, and the resulting soft supervision is combined with the original image-report supervision through a weighted hybrid objective.

We examine two pseudo-positive selection strategies: a fixed top-K approach, which selects a predefined number of reports for each sample, and a similarity-threshold approach, which allows the number of selected reports to vary according to their similarity values. Unlike Ko and Park, our implementation relies only on report-embedding similarity and does not incorporate generated negations, extracted clinical-label vectors, or medical knowledge graphs. Within this simplified setting, we evaluate the choice of sentence encoder and the soft-loss weight ($\beta$), neither of which is directly examined through comparable ablations in their work. This setting allows us to isolate how report representation, pseudo-positive selection, and soft-loss weighting affect study-level image-text retrieval.

# 3 Methods

## 3.1 Data

We utilize the MIMIC-CXR dataset [11], a large-scale multimodal corpus of chest radiographs paired with radiology reports, accessed via Kaggle. The data consists of two aligned modalities: high-resolution X-ray images and semi-structured clinical text, enabling vision-language representation learning.

The Kaggle-hosted distribution is partitioned into a training split containing approximately 368,960 images from 222,758 radiographic studies on 64,586 patients, and a validation split consisting of 2,991 images from 1,808 studies on 500 patients.

The primary variables derived from each clinical study include:

- `subject_id`: A unique patient-level identifier used to trace clinical data across an individual's history.
- `study_id`: A unique identifier for a specific X-ray examination event. (As a single patient can have multiple studies over time)
- `image`: High-resolution X-ray scans. For each study, one or more views are available, primarily consisting of Posteroanterior (PA), Anteroposterior (AP), and Lateral orientations.
- `text`: Semi-structured clinical narratives compiled by radiologists. These reports are partitioned into two key components: Findings (detailed descriptive observations) and Impressions (the concise clinical summary of the study).

The dataset is organized hierarchically, where a single patient (`subject_id`) can have multiple clinical visits (`study_id`), and each visit often contains multiple radiographic views mapped to a single free-text report. This nested, one-to-many relationship between a radiology report and its corresponding images requires preprocessing to fit the training pipeline.

Our preprocessing pipeline first flattens the nested data, so that each individual image gets its own row mapped to its corresponding text. We then use regular expressions to parse the raw text and split it into two separate, clean columns: `findings_clean` and `impression_clean`. Following this preprocessing, we identified rows with missing values in both of these columns. To maintain data quality, studies with completely empty text sections were removed, resulting in the exclusion of 14,266 rows from the training split and 129 rows from the validation split.

We further verified that the corresponding image files existed, which revealed additional missing data and led to the exclusion of 105,676 of 354,694 training rows (29.8%) and 844 of 2,862 validation rows (29.5%). The final dataset comprises 249,018 training and 2,018 validation image-report pairs.

Although training data is stored at the image level for computational efficiency, evaluation is performed at the study level by grouping all images associated with the same `study_id`.

## 3.2 Baseline Implementation

For the baseline model, we implement a CLIP-style vision-language model initialized from the pre-trained `openai/clip-vit-base-patch32` model. Inputs consist of paired chest X-ray images and clinical reports from the MIMIC-CXR dataset. Images are resized to $224 \times 224$ pixels and normalized, while reports text is tokenized and truncated to a maximum length of 77 tokens. This truncation follows the original CLIP architecture, whose text encoder is limited to 77 input tokens, ensuring compatibility with the pre-trained model. Although the original study explored multiple image resolutions, we adopted the minimum resolution of $224 \times 224$ pixels due to computational resource constraints. This choice is also consistent with the input resolution expected by the selected `ViT-B/32` backbone, which was pre-trained on images of this size [8].

Training employs a study-level contrastive loss that extends the CLIP InfoNCE objective to a supervised contrastive setting, allowing multiple images and reports associated with the same `study_id` within a batch to be treated as positive pairs. Optimization is performed using AdamW with a learning rate of $5 \times 10^{-6}$ and a weight decay of 0.2. Models are trained for up to 10 epochs with a batch size of 256, using early stopping if the validation loss does not improve for two consecutive epochs.

### 3.3 Proposed Model

The model follows the architecture and pre-trained backbone as the baseline framework (Section 3.2). To incorporate soft supervision, we compute a similarity matrix over report embeddings generated by an external language model. For each image-report pair, a soft target distribution is built over the similarity row against other reports in the batch, thereby extending supervision beyond the original pairing.

Training is performed using a hybrid objective that combines the standard CLIP cross-entropy loss ($\mathcal{L}_{CE}$) with a KL-divergence term ($\mathcal{L}_{KL}$) over the soft targets. The overall loss is defined as:

$$\mathcal{L}_{total} = (1 - \beta)\mathcal{L}_{CE} + \beta\mathcal{L}_{KL}, \tag{1}$$

where $\beta \in [0, 1]$ controls the trade-off between hard supervision and soft semantic supervision.

### 3.4 Text Embedding Models for Pseudo-Positive Construction

To construct the pseudo-positive supervision signal described in Section 3.3, we generate report-level embeddings using three pre-trained language models. These models were selected to span a range of domain specialization, from general-purpose language representations to models trained specifically on clinical and radiology text, allowing us to study how the source of semantic similarity affects the quality of the pseudo-positive structure.

- **BioClinicalBERT** [12]: A BERT-base model further pre-trained on clinical notes from the MIMIC-III database. It is domain-adapted to clinical language in general but not specifically to radiology reporting style.
- **BiomedVLP-CXR-BERT-specialized** [7]: A BERT-based text encoder pre-trained specifically on chest X-ray radiology reports as part of a vision-language pre-training framework.
- **EmbeddingGemma** [13]: A 300-million-parameter text embedding model built on the Gemma architecture, trained to produce general-purpose dense text embeddings.

Before training, we evaluated each embedding source using identical-text pairs, defined as reports whose normalized texts were identical. We compared their cosine similarities with those of all non-identical report pairs, preferring embeddings that avoided assigning uniformly high similarity across the dataset. The embedding source used in later experiments (Section 4.2) was selected using this analysis, without training a separate Soft-CLIP model for each candidate.

For each embedding model, we further evaluate the effect of the textual granularity used to compute report-report similarity, comparing three configurations: the full, unsplit report text, and the two report sections extracted during preprocessing (Section 3.1), `findings_clean` and `impression_clean`.

Comparing across these configurations allows us to disentangle two factors that could affect the quality of the soft supervision signal: the effect of the language model used, and how text granularity interacts with the CLIP encoder's 77-token limit, since full reports are more likely to be truncated than the shorter findings or impression sections.

### 3.5 From Fixed to Dynamic Pseudo-Positive Selection

By default, the soft target in Section 3.3 uses the full similarity row: every other report in the batch contributes, weighted by its similarity. We also test two ways to use a more focused set of neighbours instead: *fixed top-$K$* selection, which keeps only the $K$ most similar reports for each batch, and *threshold-based* selection.

Fixed top-$K$ selection has one problem: a report with many truly similar reports and a report with almost none both get the same number of soft targets. This ignores how dense or sparse the similarity structure really is. Ko and Park [5] solve this by using a similarity threshold instead of a fixed neighbour count, and we add the same idea to our pipeline. Instead of a fixed $K$, we apply a similarity threshold $\tau$ to the similarity matrix: any pair with similarity above $\tau$ is kept and rescaled as $(s - \tau)/(1 - \tau)$, then normalized per row into a soft target distribution. This way, the number of pseudo-positives per batch changes based on the data instead of staying fixed: reports with more similar neighbours get more pseudo-positives, and reports with fewer get fewer.

### 3.6 Evaluation

Performance is evaluated on a held-out validation set using both image-to-text and text-to-image retrieval tasks. Study-level retrieval metrics include Recall@1, Recall@5, Recall@10, Median Rank, and Mean Reciprocal Rank (MRR). Recall@K measures the proportion of queries for which the correct match appears within the top-K retrieved results. Median Rank reports the median position of the correct match in the ranked retrieval list, with lower values indicating better performance. MRR computes the average reciprocal rank of the first correct result, assigning greater weight to correct matches that appear near the top of the ranking. Together, these metrics provide a comprehensive assessment of cross-modal alignment quality.

MIMIC-CXR contains many reports with near-identical wording, and a single patient can appear in the validation set more than once. To account for this, we score retrieval under four definitions of a correct match:

- **Exact Match**: the retrieved report belongs to the same study as the query (same `study_id`).
- **Clinical**: the retrieved report has the exact same text as the query's report (after normalizing case and punctuation), regardless of which study it came from. This credits a retrieval that is wrong by `study_id` but clinically identical in content.
- **Deduplicated Exact Match**: the same rule as Exact Match, but identical reports are collapsed into one candidate to avoid penalizing the model for equivalent texts.
- **Subject**: the retrieved report belongs to the same patient (same `subject_id`), regardless of visit. This metric is used as a diagnostic measure, since a high score may reflect matching based on rare patient-specific details rather than clinical content. For example, a report naming a specific implanted device may be retrieved without broader clinical understanding.

Results are reported for a single training seed (42) per configuration.

### 3.7 Compute

Each run was trained on a single RTX 6000 GPU on Ben-Gurion University's cluster. Training took approximately 7 minutes per epoch and typically stopped after 4-6 epochs using early stopping.

## 4 Results

As stated in Section 3.6, models are evaluated on study-level image-to-text and text-to-image retrieval over the validation pool (Section 3.1), using Recall@1, Recall@5, Recall@10, Median Rank, and Mean Reciprocal Rank (MRR). Unless otherwise noted, all experiments in this section use a fixed soft-loss weight of $\beta = 0.5$, the effect of varying $\beta$ is examined separately in Section 4.5.

### 4.1 Baseline Performance

Table 1 presents the study-level retrieval performance of the baseline model. The baseline achieved an R@1 of 9.66% for image-to-text retrieval and 11.25% for text-to-image retrieval. The corresponding MRR values were 0.1645 and 0.2005. Both retrieval directions performed above the random-ranking reference, with text-to-image retrieval obtaining the higher scores across the reported metrics.

### 4.2 Embedding Source Selection

To select the embedding source, we compared the mean and standard deviation of the cosine similarities among non-identical report pairs. A lower mean and wider spread were preferred to avoid assigning similarly high scores to most pairs. As shown in Table 2, BioClinicalBERT

Table 1: Study-level retrieval performance of the baseline model on the validation set.

| Task | R@1 (%) | R@5 (%) | R@10 (%) | Median Rank | MRR |
|---|---|---|---|---|---|
| Random (2018 candidates)* | 0.050 | 0.248 | 0.496 | 1009.5 | 0.0041 |
| Image-to-Text | 9.66 | 22.20 | 31.17 | 33.0 | 0.1645 |
| Text-to-Image | **11.25** | **28.54** | **39.25** | **19.0** | **0.2005** |

*Computed analytically for uniform-random ranking over $N = 2018$ candidates.

produced a mean similarity of 0.947 and a standard deviation of 0.037, while EmbeddingGemma produced a mean of 0.723 and a standard deviation of 0.094. BiomedVLP-CXR-BERT produced the lowest mean, 0.581, and the largest standard deviation, 0.127, and was therefore selected as the embedding source used throughout the remainder of this section.

Table 2: Pairwise cosine similarity statistics between report embeddings, excluding self-comparisons.

| Embedding source | Mean | Std |
|---|---|---|
| BioClinicalBERT (mean-pooled) | 0.947 | 0.037 |
| BiomedVLP-CXR-BERT (mean-pooled) | 0.581 | 0.127 |
| EmbeddingGemma (native) | 0.723 | 0.094 |

After selecting BiomedVLP-CXR-BERT, we compared three methods for extracting its report embeddings: CLS token, mean pooling, and the model's projected representation. The projected representation was included because BiomedVLP-CXR-BERT-specialized was trained using a multimodal contrastive objective that aligns projected text and image representations in a shared embedding space [7]. All three representations were evaluated using the same similarity-based analysis.

Table 3: Pairwise cosine similarity statistics for BiomedVLP-CXR-BERT, by pooling method.

| Pooling method | Mean | Std |
|---|---|---|
| CLS | 0.468 | 0.197 |
| Mean | 0.581 | 0.127 |
| Projection | 0.035 | 0.395 |

As shown in Table 3, the projected representation produced the lowest mean similarity, 0.035, and the largest standard deviation, 0.395. It was therefore used in the remaining experiments.

### 4.3 Effect of Text Granularity on Retrieval

This experiment examines whether the text granularity used to construct the pseudo-positive similarity matrix described in Section 3.3 affects retrieval performance. All other settings are fixed, and the soft target is constructed from the full batch without a top-K or similarity-threshold restriction. The model receives the full report text in every run; only the text used to calculate report-report similarities changes. We compare three text granularities: the full, unsplit report, findings_clean, and impression_clean, as introduced in Section 3.1. A separate Soft-CLIP model is trained for each configuration and evaluated using the study-level protocol described in Section 3.6.

Table 4 presents the retrieval results for both directions and includes the hard CLIP baseline, which does not use a soft target. Differences among the three granularities are small, with no consistent winner. For image-to-text retrieval, all Soft-CLIP configurations improve R@K over the baseline, with the full report and impression_clean generally performing best. For text-to-image retrieval, the full report is the most consistent; findings_clean also improves all metrics, while impression_clean improves all except R@1, although both lead on individual metrics.

Overall, no granularity is consistently best across metrics and retrieval directions, and the differences are generally small (e.g., under 1 point on R@1 and R@10 for both tasks). Although the comparison was motivated by the CLIP text encoder's 77-token truncation limit, the shorter sections show no

Table 4: Retrieval performance by pseudo-positive text granularity.

| Task | Granularity | R@1 (%) | R@5 (%) | R@10 (%) | Median Rank ↓ | MRR ↑ |
|---|---|---|---|---|---|---|
| I2T | Hard CLIP baseline | 9.66 | 22.20 | 31.17 | 33.0 | 0.1645 |
| | Full report text | 10.85 | **24.18** | 32.46 | 30.0 | **0.1800** |
| | findings_clean | 10.41 | 22.45 | 31.32 | 33.0 | 0.1690 |
| | impression_clean | **11.10** | 23.64 | **33.20** | **29.0** | 0.1791 |
| T2I | Hard CLIP baseline | 11.25 | 28.54 | 39.25 | 19.0 | 0.2005 |
| | Full report text | **12.24** | 30.43 | 41.58 | **16.0** | **0.2158** |
| | findings_clean | 11.60 | **30.82** | 41.28 | 18.0 | 0.2135 |
| | impression_clean | 11.15 | 30.08 | **41.87** | 17.0 | 0.2090 |

clear advantage over full reports. We therefore use the full report text in subsequent experiments, maintaining consistency with the embedding comparison in Table 2.

### 4.4 Pseudo-Positive Selection

We compared three pseudo-positive selection approaches: using the full batch, selecting the top-K similar reports, and selecting reports above a similarity threshold $\tau$.

Thresholds at the 90th, 95th, and 97th percentiles were calculated from 5,000 training reports, giving $\tau \in \{0.681, 0.828, 0.894\}$. They retain approximately 10%, 5%, and 3% of pairs per row and provide increasingly strict pseudo-positive selection while remaining below the near-duplicate region described in Section 4.2. The corresponding similarity distribution histogram is provided in Appendix A.

Table 5 presents the threshold-based results. Across both retrieval tasks, $\tau = 0.894$ leads on the majority of metrics, achieving the best R@5, R@10, Median Rank, and MRR for image-to-text and the best R@1, R@5, R@10, and MRR for text-to-image. $\tau = 0.828$ is competitive only on image-to-text R@1 and Median Rank, while $\tau = 0.681$ does not lead on any metric. This indicates that the stricter, more selective threshold provides the most consistent improvement over the full-batch configuration.

Table 5: Retrieval performance using full-batch and threshold-based selection.

| Task | Configuration | R@1 (%) | R@5 (%) | R@10 (%) | Median Rank | MRR |
|---|---|---|---|---|---|---|
| I2T | Full batch | 10.85 | 24.18 | 32.46 | 30.0 | 0.1800 |
| | $\tau = 0.681$ (p90) | 11.55 | 23.49 | 32.21 | 31.5 | 0.1826 |
| | $\tau = 0.828$ (p95) | **11.69** | 24.18 | 32.71 | 29.0 | 0.1860 |
| | $\tau = 0.894$ (p97) | 11.60 | **24.68** | **33.94** | **28.0** | **0.1867** |
| T2I | Full batch | 12.24 | 30.43 | 41.58 | **16.0** | 0.2158 |
| | $\tau = 0.681$ (p90) | 12.09 | 29.19 | 41.43 | 17.0 | 0.2151 |
| | $\tau = 0.828$ (p95) | 10.56 | 30.82 | 42.32 | 16.5 | 0.2080 |
| | $\tau = 0.894$ (p97) | **12.54** | **31.96** | **43.95** | 17.0 | **0.2236** |

Following the same setup as the threshold experiments, we evaluate fixed top-$K$ pseudo-positive selection with $K \in \{5, 10, 20\}$, alongside the full-batch configuration.

Table 6 presents the fixed top-$K$ results. For image-to-text retrieval, $K = 10$ achieves the best R@1 and MRR, while $K = 5$ achieves the best R@5, R@10, and Median Rank; $K = 20$ does not lead on any metric. For text-to-image retrieval, no single configuration is consistently best: $K = 20$ achieves the highest R@1 and MRR, $K = 10$ the highest R@5, and the full-batch configuration the highest R@10, with Median Rank tied across all configurations. This suggests that, unlike the threshold-based selection, fixed top-$K$ does not yield a clearly preferable value of $K$ across both retrieval directions.

Table 6: Retrieval performance using fixed top-$K$ selection.

| Task | Configuration | R@1 (%) | R@5 (%) | R@10 (%) | Median Rank | MRR |
|---|---|---|---|---|---|---|
| I2T | Full batch | 11.65 | 24.58 | 33.50 | 28.0 | 0.1869 |
| | $K = 5$ | 11.79 | **25.17** | **34.39** | **27.5** | 0.1880 |
| | $K = 10$ | **12.29** | 25.07 | 33.94 | **27.5** | **0.1903** |
| | $K = 20$ | 11.89 | 24.48 | 33.40 | 29.0 | 0.1868 |
| T2I | Full batch | 11.00 | 31.67 | **43.76** | 16.0 | 0.2145 |
| | $K = 5$ | 11.20 | 30.53 | 43.46 | 16.0 | 0.2147 |
| | $K = 10$ | 11.05 | **31.96** | 42.52 | 16.0 | 0.2136 |
| | $K = 20$ | **12.29** | 31.52 | 43.71 | 16.0 | **0.2205** |

### 4.5 Soft-Loss Weight Sensitivity

As described in Section 3.3, $\beta$ controls the trade-off between the hard and soft supervision terms in Equation 1. All previous experiments in this section used a fixed value of $\beta = 0.5$; here, we vary $\beta \in \{0.0, 0.1, 0.3, 0.5, 0.7, 1.0\}$ while keeping the embedding source, text granularity, and pseudo-positive selection strategy fixed to the settings selected above. $\beta = 0.0$ recovers the hard-CLIP baseline, and $\beta = 1.0$ relies entirely on the soft KL term.

Table 7 presents the $\beta$-sensitivity results. $\beta = 0.3$ dominates across both retrieval tasks, achieving best performance on every metric for image-to-text and the best R@1, R@5, and Median Rank for text-to-image, while remaining close to the best on the R@10. Both smaller ($\beta \leq 0.1$) and larger ($\beta \geq 0.5$) values underperform this setting, and $\beta = 1.0$, which relies solely on the soft KL term, performs the worst or near-worst on nearly every metric in both tasks. This indicates that soft supervision works best as a moderate complement to the hard loss, not a substitute: too little soft weight adds negligible signal, while removing the hard term removes the direct incentive to align each image with its own report.

Table 7: Retrieval performance by $\beta$.

| Task | Configuration | R@1 (%) | R@5 (%) | R@10 (%) | Median Rank | MRR |
|---|---|---|---|---|---|---|
| I2T | $\beta = 0.0$ (hard baseline) | 9.66 | 22.20 | 31.17 | 33.0 | 0.1645 |
| | $\beta = 0.1$ | 10.31 | 22.05 | 31.76 | 32.0 | 0.1687 |
| | $\beta = 0.3$ | **11.94** | **25.47** | **34.74** | **27.0** | **0.1900** |
| | $\beta = 0.5$ | 11.65 | 24.58 | 33.50 | 28.0 | 0.1869 |
| | $\beta = 0.7$ | 10.65 | 23.79 | 32.76 | 32.0 | 0.1773 |
| | $\beta = 1.0$ | 10.51 | 23.04 | 31.17 | 37.0 | 0.1718 |
| T2I | $\beta = 0.0$ (hard baseline) | 11.25 | 28.54 | 39.25 | 19.0 | 0.2005 |
| | $\beta = 0.1$ | 11.00 | 28.89 | 39.15 | 19.0 | 0.2015 |
| | $\beta = 0.3$ | **12.14** | **32.66** | 42.81 | **15.0** | **0.2225** |
| | $\beta = 0.5$ | 11.00 | 31.67 | **43.76** | 16.0 | 0.2145 |
| | $\beta = 0.7$ | 10.60 | 30.43 | 41.72 | 17.0 | 0.2073 |
| | $\beta = 1.0$ | 11.45 | 29.04 | 37.91 | 22.0 | 0.2034 |

### 4.6 Combined Configuration

Building on the results of Sections 4.2–4.5, we train a final model combining the best-performing setting from each ablation: BiomedVLP-CXR-BERT projected embeddings over the full report text (Section 4.3), threshold-based pseudo-positive selection with $\tau = 0.894$ (Section 4.4), and $\beta = 0.3$ (Section 4.5). Table 8 reports its performance under all four match definitions introduced in Section 3.6.

Exact Match and Clinical scores are nearly identical for both tasks, indicating that near-duplicate reports rarely change the retrieved study's rank. Deduplicated Exact Match improves every metric over Exact Match in both directions (e.g., I2T R@10 rises from 34.54% to 41.97%), confirming that competing near-duplicate candidates in the pool depress the raw study-level score. The Subject definition yields substantially higher recall and lower Median Rank than the other three (e.g., I2T R@1 of 30.82% versus 11.65%), consistent with the diagnostic role assigned to this metric in Section 3.6: it is more permissive by construction and likely reflects patient-specific retrieval cues rather than a genuine improvement in clinical alignment.

Table 8: Retrieval performance of the combined configuration ($\tau = 0.894$, $\beta = 0.3$, BiomedVLP-CXR-BERT projected embeddings, full report text) under all four match definitions.

| Task | Match definition | R@1 (%) | R@5 (%) | R@10 (%) | Median Rank | MRR |
|---|---|---|---|---|---|---|
| I2T | Exact Match | 11.65 | 24.83 | 34.54 | 28.0 | 0.1864 |
| | Clinical | 11.65 | 24.83 | 34.54 | 28.0 | 0.1864 |
| | Deduplicated Exact Match | 11.65 | 30.62 | 41.97 | 17.0 | 0.2141 |
| | Subject | **30.82** | **49.45** | **59.66** | **6.0** | **0.3968** |
| T2I | Exact Match | 11.55 | 31.07 | 42.12 | 15.0 | 0.2180 |
| | Clinical | 11.55 | 31.47 | 42.52 | 15.0 | 0.2198 |
| | Deduplicated Exact Match | 13.43 | 34.89 | 46.23 | 13.0 | 0.2424 |
| | Subject | **26.61** | **50.69** | **60.56** | **5.0** | **0.3799** |

## 5 Discussion

The most notable finding to emerge from our experiments is not any single ablation result, but a pattern that cuts across all of them: under the Subject match definition, retrieval performance is substantially higher than under any of the three study-level definitions (Exact Match, Clinical, and Deduplicated Exact Match). For the combined configuration (Section 4.6), image-to-text R@1 under Subject reaches 30.82%, nearly three times the 11.65% obtained under Exact Match, with a similar gap in the text-to-image direction and a correspondingly lower Median Rank. This gap is far larger than any difference observed between embedding sources, text granularities, selection strategies, or values of $\beta$ throughout Section 4.

This gap suggests that the model is partly relying on patient-identifying signal rather than clinical content to perform retrieval. As noted in Section 3.6, the Subject definition credits any retrieval from the same patient, regardless of visit, so a high score here can come from recognizing something that marks a specific patient rather than a specific finding. A plausible example is an implant: a pacemaker or other device visible on the X-ray would look nearly identical across all of a patient's studies, regardless of what else changed clinically between visits. Another is surgical hardware - a patient with a documented history of chest surgery may show sternal wires from that procedure in every subsequent study, years apart, giving the model a stable visual anchor tied to the patient rather than to the report content.

The retrieval shown in Figure 1 offers a plausible illustration of this type of shortcut. The query image shows a new finding in the lower left lung that was interpreted as pneumonia. However, the retrieved report comes from a different study of the same patient and states that no acute problem was found. Both images show median sternotomy wires from a prior cardiac procedure, a permanent and visually distinctive feature that remains visible across the patient's studies but is unrelated to the new finding. The model may therefore be using these wires as a stable patient-specific cue rather than matching the current clinical finding. Additional examples are provided in Appendix B.

This raises a validity concern as some of the improvements we report under the primary metrics (Exact Match, Clinical) could be partly driven by the same patient-level shortcuts rather than purely clinical alignment. It also points to a mild privacy consideration: a model that can identify which patient a chest X-ray belongs to, even unintentionally, touches on re-identification concerns worth flagging, even though mitigating them is outside the scope of this project.

We also checked whether this pattern held across patient subgroups. Splitting the validation set by number of studies per patient, image-to-text R@1 under Exact Match shows no significant gap between single-study and multi-study patients (+1.62pp, 95% CI [-3.02, +6.06]), while under Subject the gap is large and clearly significant (+24.69pp, 95% CI [+16.67, +32.64]). This suggests the shortcut we identified is tied specifically to how the Subject metric is defined, rather than one that clearly carries over to our primary clinical metrics.

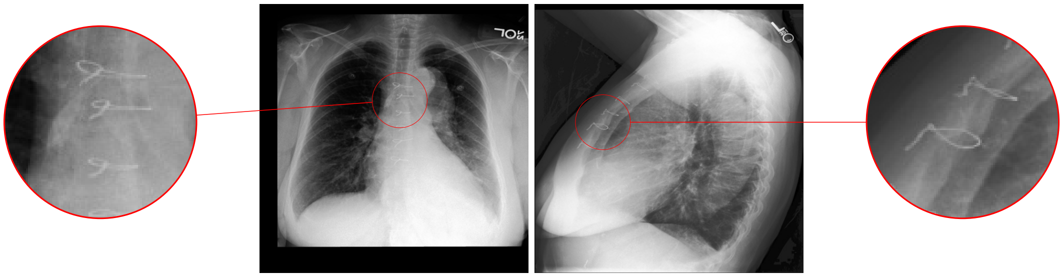

Figure 1: Query image (left, study 51697632) and retrieved image (right, study 54070533). Both show median sternotomy wires despite differing in projection and in reported finding.

Alongside this concern, threshold-based selection produced the most reliable gain over the hard-CLIP baseline: the stricter threshold ($\tau = 0.894$) led on nearly every metric in both retrieval directions (Section 4.4), unlike fixed top-$K$ selection, which showed no consistent preferred value of $K$. The soft-loss weight $\beta$ showed the same pattern from the opposite side: $\beta = 0.3$ clearly outperformed all other values we tested, while $\beta = 1.0$, which discards the hard loss entirely, performed worst or near-worst on almost every metric (Section 4.5). Taken together, these results show that soft supervision helps only when it complements rather than replaces direct image-report alignment, and that it is most effective when the number of pseudo-positives reflects how similar the neighbours actually are, rather than being fixed in advance. The fact that full-text, findings, and impression embeddings perform similarly indicates that truncation does not meaningfully cost the full-text embedding useful signal, and that findings and impressions capture complementary rather than redundant information.

A further limitation concerns how the embedding source itself was selected. As described in Section 4.2, we chose among the three candidate embedding models using a proxy criterion - the mean and spread of cosine similarity between non-identical reports - rather than by training a full Soft-CLIP model for each candidate and comparing retrieval performance directly, as we did for our other hyperparameters. A more discriminative similarity distribution is a reasonable proxy for producing a useful pseudo-positive structure, but it is not guaranteed to translate into better downstream retrieval, and a less "clean" embedding source could in principle still yield stronger soft targets in practice. A related limitation is that all configurations were trained and evaluated with a single random seed, so small differences between neighbouring threshold or $K$ values may reflect run-to-run variance.

Finally, as this was our first project working with medical data, and neither of us has a clinical background, we approached the domain-specific aspects of this work with appropriate caution and did our best throughout. We come away with valuable lessons for future work with medical data.

## 6 Conclusion

Standard CLIP training treats every non-matching image-report pair as an equally hard negative, even though radiology reports frequently share clinical content. We addressed this by augmenting CLIP with a soft KL term built from report-embedding similarity, comparing fixed top-$K$ and similarity-threshold pseudo-positive selection across several embedding sources and soft-loss weights. Our best configuration improved over the hard-CLIP baseline in both retrieval tasks, with threshold-based selection proving more reliable than fixed top-$K$, and soft supervision helping as a moderate complement to the hard loss.

Alongside these gains, our four-way evaluation protocol surfaced a shortcut: under the Subject match definition, the model scores far higher than under any study-level definition, and qualitative examples (Section 5, Appendix B) show retrievals apparently anchored to patient-specific artifacts, rather than clinical content. A subgroup analysis suggests this shortcut is largely confined to the Subject metric itself rather than inflating our primary results, but it underscores the need for match definitions that separate clinical similarity from patient identity when evaluating retrieval on datasets with repeated patients. Future work should evaluate the embedding source end-to-end rather than via a proxy criterion, repeat these experiments across multiple seeds, and probe the patient-identity shortcut more directly, for example by penalizing retrievals that rely on non-clinical visual cues.

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## A Similarity Distribution and Threshold Percentiles

Figure 2 shows the off-diagonal cosine similarity distribution among projected BiomedVLP-CXR-BERT report embeddings (train split), with the evaluated threshold percentiles marked.

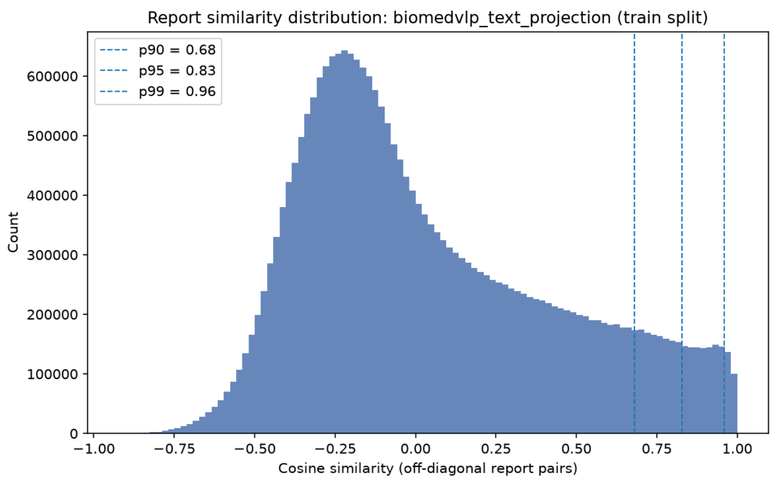


Figure 2: Off-diagonal cosine similarity distribution (train split, biomedvlp_text_projection), with evaluated threshold percentiles marked.

## B Additional Patient-Identity Retrieval Examples

This appendix provides two additional examples of the pattern discussed in Section 5: retrieval succeeds under the Subject definition but fails under Exact Match and Clinical, with the retrieved item's content not accounting for the match. In both cases, the true and retrieved items belong to the same patient. As discussed below, the specific feature driving the retrieval cannot be identified with confidence.

### Image-to-Text

In the image-to-text direction the query is the radiograph itself (Figure 3), and the model must return the report that belongs to it:

> **True report (Study 53275640):** *"Findings:* ***The tracheostomy tube is unchanged in position*** *and terminates approximately 4.8 cm above the carina. The right PICC line terminates in the distal SVC.* ***There is no significant change in the lungs*** *when compared to ___. There are several parenchymal calcifications which were characterized on the most recent CT scan. Again noted are diffuse infiltrative parenchymal opacities, right worse than left; this is largely due to pulmonary edema and the right-sided pleural effusion, but underlying pneumonia cannot be excluded. The mediastinum is wide, which was noted as far back as the outside hospital CXR from ___. No acute osseous abnormalities. Impression: 1.* ***Moderate pulmonary edema, unchanged.*** *2.* ***Interval improvement in right-sided pleural effusion."***

The model instead returned a report from a different study of the same patient:

> **Retrieved report (Study 50880103):** *"Findings: on the **chest CT** ___ showed probable multi focal pneumonia, predominantly in the right lung, and mild interstitial edema. Edema improved between ___ and ___, and then opacification in the right lung increased again accompanied by increasing moderate right pleural effusion. The progression of these associated findings this suggested that the interval change was primarily due to cardiac decompensation. **Today edema has worsened in both lungs**, and **the moderate right pleural effusion is larger**, although the opacification in the left lower lung is heterogeneous enough to suggest concurrent pneumonia or large scale aspiration. Mild cardiomegaly and chronic mediastinal widening are chronic."* (Impression repeats the Findings text almost verbatim.)

In plain terms, the true report says the breathing tube and IV line sit where they should, the lungs are **essentially unchanged**, and the fluid around the right lung has **improved**. The retrieved report tells a different, multi-day story built around a CT scan the true report never mentions, in which fluid has **worsened in both lungs** and the collection around the right lung is **larger**; it never mentions the breathing tube or the IV line at all. The two reports do not describe the same picture, so the retrieved report's content does not explain the match.

However, unlike the sternotomy-wire example discussed in Section 5, this case does not contain a single clearly identifiable feature that could explain the same-patient retrieval. The query image (Figure 3) is a radiograph with some rotation and clutter. Although these characteristics may have contributed to the match, none is sufficiently distinctive to support a confident explanation. Nevertheless, the clear mismatch between the image and the retrieved report, although ambiguous, suggests that the model relied on information other than the current clinical findings.


Figure 3: Query image (study 53275640).

**Text-to-Image**

A similar pattern appears in the text-to-image direction, where the query is the report itself:

> **Query report (Study 50391562):** *"Findings: Impression: **Small to moderate right subpulmonic pleural effusion has re accumulated**, substantially smaller than its volume on ___. Aside from mild right basal atelectasis lungs are clear. There is no left pleural effusion. There is no evidence of central lymph node enlargement. Incidental note is made of a **heavily calcified mitral anulus** and possible left atrial enlargement, but there is no overall cardiomegaly or any pulmonary vascular congestion or pulmonary edema."*

In plain terms, the query describes fluid around the right lung that has **returned** since a previous study, and notes in passing a long-standing calcium deposit on a heart valve. The retrieved image (Figure 4) comes from a different study of the same patient, and its report states that the lungs are clear and that *"there is* ***no definitive pleural effusion*** *seen"* (Study 56219888). By the radiologists' own readings, the retrieved image therefore does not show the finding the query describes.

That same report also notes a *"calcified mitral anulus,"* the same incidental detail described in the query and a chronic feature of this patient unrelated to whether fluid is present at a given visit. Its appearance in both reports suggests that a stable, patient-specific finding links the two studies and may help explain the match, although the finding itself is too subtle to confirm in the radiographs.

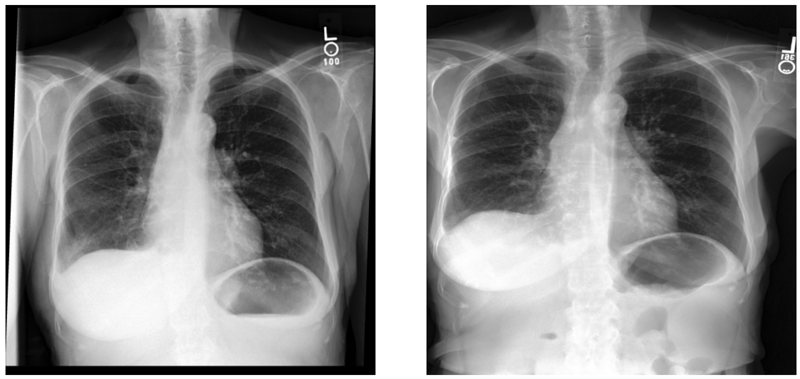

Figure 4: True image (left, study 54197091) and retrieved image (right, study 54296756).

Taken together, these examples suggest that patient identity may influence retrieval beyond clinical similarity alone. In both directions, the top-ranked result came from the correct patient despite describing a different clinical state. This pattern indicates that the model may encode stable, patient-specific characteristics in addition to the findings present in each examination.